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Introduction to Probability

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Probability

Probability is the Science of decision-making with calculated risks in the face of uncertainty.

--Ya Lin Chou

Introduction

- Probability is a fundamental concept in mathematics that quantifies the likelihood of an event occurring.
- Probability provides a numerical way to express the chance of a random event happening.
- In simpler terms, it's about predicting how likely something is to happen.
- Its applications are vast and touch almost every aspect of modern life, from scientific research to everyday decision-making.

Applications of Probability

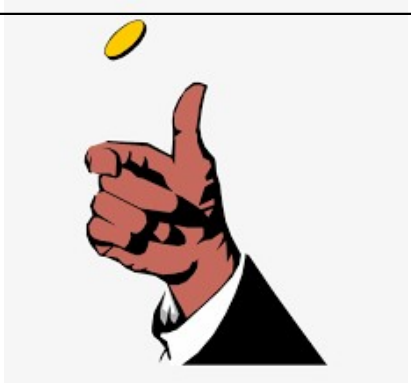
Some key areas where probability is extensively used are:

- Statistics and Data Analysis
- Finance and Economics
- Science and Engineering
- Everyday Life and Decision Making
- Computer Science and Artificial Intelligence
- Social Sciences

Terminology

Experiment (or) Trial: A process or activity that has a well-defined set of possible outcomes.

Examples:

Flipping a Coin	Rolling a die	Drawing a card from a deck.
		

Terminology

Random Experiment: A Random experiment is an experiment whose outcomes can't be predicted with certainty. The outcome is not unique but may be any one of the possible outcomes.

Examples:

Flipping a Coin	Rolling a die	Drawing a card from a deck.
		
Head or Tail	**1,2,3,4,5, or 6**	**Any card from the deck of 52**

Terminology

Sample Space: The set of all possible outcomes of an experiment is known as sample space and it is denoted by S or Ω.

Examples:

Experiment	Possible Outcomes	Sample Space
Tossing a Coin	Head or Tail	S = {H,T}
Throwing die	1,2,3,4,5,6	S = {1, 2, 3, 4, 5, 6}
Playing a Game	Win or Lose	S={W,L}

Sample Space

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EXPERIMENT

- Consider an experiment that selects a cell phone camera and records the recycle time of a flash (the time taken to ready the camera for another flash).
- The possible values for this time depend on the resolution of the timer and on the minimum and maximum recycle times.
- However, because the time is positive, it is convenient to define the sample space as simply the positive real line

$$S = \Omega = R^{+} = \{X/x > 0\}$$

Note: X is considered as the 'recycle time of a flash'.



Sample Space

Research Questions: Identify the sample space, if

1. all the recycle times are between 1.5 and 5 seconds
2. the objective of the analysis is to consider only whether the recycle time is low, medium, or high
3. the objective is only to evaluate whether a particular camera conforms to a minimum recycle-time specification

Sample Space (Solutions)

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1. If it is known that all recycle times are between 1.5 and 5 seconds, the sample space can be

$$\boldsymbol{S = \Omega = R^{+} = \{X/1.5 < x < 5\}}$$

2. If the objective of the analysis is to consider only whether the recycle time is low, medium, or high, the sample space can be taken to be the set of three outcomes

$$\boldsymbol{S = \Omega = \{Low,\ Medium,\ High\}}$$

3. If the objective is only to evaluate whether or not a particular camera conforms to a minimum recycle-time specification, the sample space can be simplified to a set of two outcomes:

$$\boldsymbol{S = \Omega = \{Yes, No\}}$$

Types of Sample Space

Sample Space	Definition	Examples
Discrete	A sample space is discrete if it consists of a finite or countably infinite set of outcomes.	$S = \{Low, Medium, High\}$ $S = \{Yes, No\}$
Continuous	A sample space is continuous if it contains an interval (either finite or infinite) of real numbers.	$S = R^{+} = \{X/x > 0\}$ $S = R^{+} = \{X/1.5 < x < 5\}$

Events and Types of Events

Type of Event	Definition	Example
Event	The outcome, or combination of outcomes, of an experiment is termed as an event.	• Tossing a coin is a random experiment, and getting a head or a tail is an event. • Throwing a die is a random experiment, and getting numbers 1,2,…,6 is called an events.
Simple Event	A single possible outcome of an experiment is called a simple event.	• If we toss an unbiased coin, then the single outcome 'Head' or 'Tail' is known as a simple event. • Symbolically, if we are tossing a single coin the sample space is, S={H,T} E_1 = The event of getting a head E_2 = The event of getting a Tail are called the two simple events of the experiment.

Types of Events

Type of Event	Definition	Example
Null Event	Null event contains no sample point . Null event is also known as impossible event. It is usually denoted by, $\emptyset = \{\}$.	In tossing a coin, the event that the coin will stand erect on its edge is a null event because there is no outcome in this situation.
Mutually Exclusive Events	• If two or more events can't occur simultaneously in a single trail of an experiment, then such events are called mutually exclusive events or disjoint events. • If E_1 *and* E_2 are any two mutually exclusive events, then $E_1 \cap E_2 = \phi$	**Example-1:** If we toss a coin the events of getting H and T can't occur simultaneously. **Example-2**: If we are throwing a die the events of getting numbers 1,2,…,6 can't occur simultaneously.

Types of Events

Type of Event	Definition	Example
Exhaustive Events	The total number of possible outcomes of a random experiment is known as the exhaustive events.	If we toss an unbiased coin, then the two events are: E_1 = The event of getting a head E_2 = The event of getting a Tail are called the exhaustive events, because there are no other events in this experiment.
Equally Likely Events	Two or more events are said to be equally likely when each of these has an equal chance to occur.	**Example-1**: In a random toss of an unbiased coin, head and tail are equally likely events. **Example-2**: In throwing an unbiased die, all the six faces are equally likely events.
Independent Events	Two events can't be dependent on each other	If we toss an unbiased coin, then the two events: E_1 = The event of getting a head E_2 = The event of getting a Tail are independent of each other.

Mathematical (or) Classical definition of Probability

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Definition:

If a random experiment results 'n' outcomes, out of which outcomes 'm' are in favor of event E and all these outcomes are exhaustive, mutually exclusive and equally likely, then the probability of occurrence of event E is defined as:

$$P(E) = \frac{Favourable\ number\ of\ outcomes}{Total\ Number\ of\ outcomes}$$

$$P(E) = \frac{m}{n}$$

Limits of Probability

Since from the definition, $m \geq 0,\ n \geq 0 \quad and \quad m \leq n$, we have

$$P(E) = \frac{m}{n} \geq 0$$

$$P(E) = \frac{m}{n} \leq 1$$

$$\Rightarrow \quad 0 \leq P(E) \leq 1$$

Therefore, the probability of any event lies between 0 and 1.

Properties

- The probability of an event E can be represented in terms of the odds.
- From the definition of probability, the odds in favour of E are **m : (n-m),** and the odds against E are **(n-m) : m.**
- For example, let us assume that, when we toss a coin ten times, the odds in favour of the event of getting a head are 4:6.
- It means, out of 10 times, the favorable number of times we can get heads is 4 times, and it is possible to get tails 6 times.

  i. The probability of getting head= $P(E) = \frac{4}{10}$

  ii. The probability of getting tail = $P(E) = \frac{6}{10}$

Properties

- The non-happening of the event E is called the complementary event of E and is denoted by $\overline{E}$ *or* E^c.

$$P(E) + P(\overline{E}) = 1$$

- The probability 'p' of the happening of an event is also known as the probability of success and the probability 'q' of the non-happening of the event as the probability of failure.

$$p + q = 1$$

- If $P(E) = 1$, E is called a certain event.
- If $P(E) = 0$, E is called an impossible event.

Statistical (or) Empirical Probability

Definition:

If an experiment is performed repeatedly under essentially homogeneous and identical conditions, then the limiting value of the ratio of the number of times the event occurs to the number of trials, as the number of trials becomes indefinitely large, is called the probability of happening of the event, it being assumed that the limit is finite and unique.

Symbolically, if in 'n' trials and event 'E' happens 'm' times, then the probability of the happening of 'E', denoted by $P(E)$ and is given by

$$P(E) = \lim_{n \to \infty} \frac{m}{n}$$

Statistical (or) Empirical Probability

Note:

J.E Kerrich conducted coin tossing experiment with 10 sets of 1,000 tosses each during his confinement in World War-II. The number of heads found by him were:

$$502,\ 511,\ 497,\ 529,\ 504,\ 476,\ 507,\ 520,\ 504,\ 529$$

From this empirical observations, the probability of getting head in a toss of an unbiased coin is : $\frac{5079}{10{,}000} = 0.5079 \cong \frac{1}{2}$.

Thus, the empirical probability approaches the classical probability as the number of trials becomes indefinitely large.

Axiomatic Definition of Probability

Let S be the sample space of an experiment. Probability measure P(.) is a real-valued function that assigns to each event A in the sample space S, a number P(A), called the probability of A, with the following conditions satisfied:

i. It is nonnegative, $P(A) \geq 0$

ii. It is unity for a certain event. That is, $P(S) = 1$

iii. It is additive over the union of an infinite number of pairwise disjoint events. Symbolically, if $A_1, A_2, \ldots$ form a sequence of pairwise mutually exclusive events in S, then

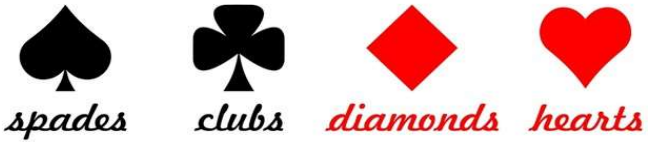
$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

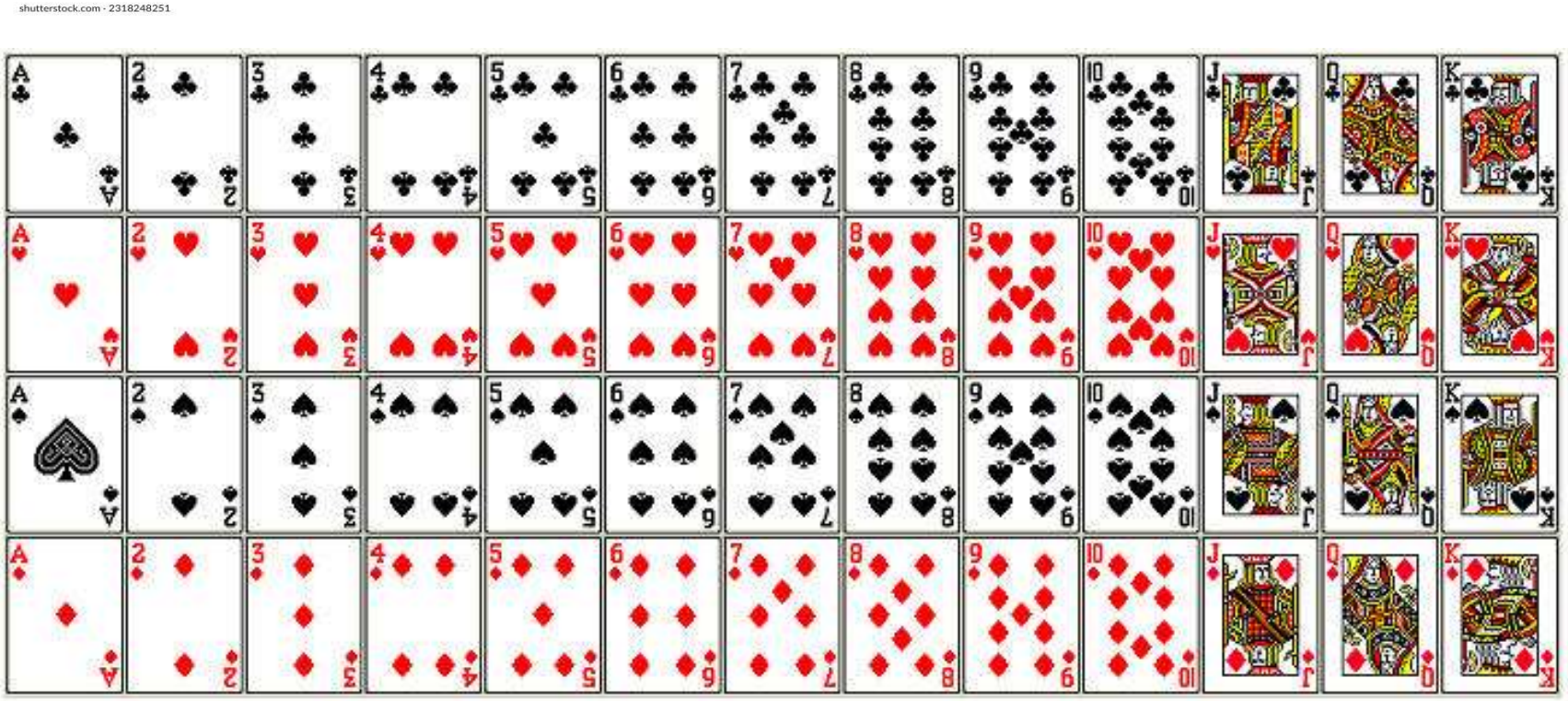
Addition Law of Probability

	Types of Event	**Definition**
Definition-I	Non-Disjoint events	If A and B are any two non-disjoint events, we have $P(A \cap B) \neq 0$, then the addition theorem of probability is given by: $$\boldsymbol{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$
Definition-II	Disjoint Events	If A and B are disjoint events, we have $P(A \cap B) = 0$, then the addition theorem of probability is given by: $$\boldsymbol{P(A \cup B) = P(A) + P(B)}$$
Definition-III	Independent Events	If A and B are any two independent events, then we have: $$\boldsymbol{P(A \cap B) = P(A).P(B)}$$ The addition theorem of probability can be written as $$\boldsymbol{P(A \cup B) = P(A) + P(B) - P(A).P(B)}$$

Playing Cards

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Examples

Example-1:

In a large metropolitan area, the probabilities of having a color television set, an HDTV set, and both a color television and an HDTV set are 0.86, 0.35, and 0.29, respectively. What is the probability that a family owns either or both kinds of sets?

Example-2:

A card is drawn from a well-shuffled pack of playing cards. What is the probability that it is either a spade or an ace? (Ans: 16/52).

Example-3:

The odds against A solving a certain problem are 4 to 3, and the odds in favor of B solving the same problem are 7 to 5. What is the probability that the problem is solved if both try independently? (Ans. 0.7619).

Examples

Example-4:

A committee of size 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

Example-5:

A box contains 6 red, 4 white, and 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn, there is at least one ball of each color. (Ans: 0.5275).

Conditional Probability

Definition-1	If A and B are any two events, then the conditional probability of A given B is $$P(A/B) = \frac{P(A \cap B)}{P(B)}\ , P(B) > 0$$ Here $P(A/B)$ represents the conditional probability of occurrence of A when the event B is already happened.
Definition-2	If A and B are any two events, then the conditional probability of B given A is $$P(B/A) = \frac{P(B \cap A)}{P(A)}\ , P(A) > 0$$ Here $P(B/A)$ represents the conditional probability of occurrence of B when the event A is already happened.

Multiplication Law of Probability

	Types of Event	Definition
Definition-I	The two events A and B are dependent	If A and B are any two events, then the multiplication theorem of probability is defined as: $$P(A \cap B) = \begin{cases} P(A)\ P(B/A), & P(A) > 0 \\ P(B)P(A/B), & P(B) > 0 \end{cases}$$ Here $P(A/B)$ and $P(B/A)$ are the conditional probabilities.
Definition-II	The two events A and B are independent	If A and B are independent, then we have $$P(A/B) = P(A), \text{ and } P(B/A) = P(B)$$ The multiplication law of probability becomes as $$P(A \cap B) = P(A)\ P(B)$$

Examples

Example-6:

A manufacturer of airplane parts knows from past experience that the probability is 0.80 that an order will be ready for shipment on time, and it is 0.72 that an order is ready for shipment on time and will be delivered on time. What is the probability that such an order will be delivered on time, given that it was ready for shipment on time?. (Ans: 0.90).

Example-7:

Find the probabilities of randomly drawing two aces in succession from an ordinary deck of 52 playing cards if we sample

(i) Without replacement

(ii) With replacement

Bayes Theorem

Definition: If $E_1, E_2, \ldots, E_n$ are mutually disjoint events with $P(E_i) \neq 0$, then for any arbitrary event A, which is a subset of $\bigcup_{i=1}^{n} E_i$, such that $P(A) > 0$, we have

$$P\left(E_i/A\right) = \frac{P(E_i)\, P\left(A/E_i\right)}{\sum_{i=1}^{n} P(E_i)\, P\left(A/E_i\right)},\ \mathrm{i} = 1,2,\ldots,\mathrm{n}$$

Here

(i) $P(E_i)$ are 'a priori probabilities

(ii) $P\left(A/E_i\right)$ are likelihoods

(iii) $P\left(E_i/A\right)$ are called 'posterior probabilities

Bayes Theorem

Example-1:

A box contains 750 IC chips of which 485 are manufactured by company A and the rest by company B. It is estimated that 5% of the chips made by company A and 2% made by company B are defective. If a randomly selected chip is found to be defective, find the probability that it is manufactured by company A.

Bayes Theorem

Solution:

Let,

$E_1, \ and \ E_2$ denote the events that IC chips manufactured by company A and B.

$$P(E_1) = \frac{485}{750} = 0.65, \ and \quad P(E_2) = \frac{265}{750} = 0.35$$

Let,

E denote the event that 'the selected IC chip being defective'.

$$P(E/E_1) = 5\% = \frac{5}{100} = 0.05, \quad P(E/E_2) = 2\% = \frac{2}{100} = 0.02$$

Bayes Theorem

$P(E) = P$(The selected IC chip is defective)

$$P(E) = P(E_1 \cap E) + P(E_2 \cap E)$$

According to the multiplication law of probability, we have,

$$P(E) = P(E_1)P(E/E_1) + P(E_2)P(E/E_2) = 0.0395$$

Using the Bayes' theorem, the probability that 'the defective IC chip manufactured by Company A' is

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E)} = \frac{0.0325}{0.0395} = 0.8228$$

Bayes Theorem

Example-2:

In a bolt factory machines A, B and C manufacture respectively 25%, 35% and 40% of the total. Of their output 5, 4, 2 percent are defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machines A, B and C?

(Ans. 25/69, 28/69, 16/69)

Bayes Theorem

Example-3:

The probabilities of A, B and C becoming managers of a company are 4/9, 2/9, and 1/3, respectively. The probabilities that the Bonus will be introduced if A, B and C become managers are 3/10, ½ and 4/5 respectively.

(i) What is the probability that Bonus will be introduced?

(ii) If the Bonus Scheme has been introduced, what is the probability that manager appointed was A?

Bayes Theorem

Solution:

Let,

$E_1, E_2, \ and \ E_3$ denote the events that A, B and C become managers.

$$P(E_1) = \frac{4}{9}, \quad P(E_2) = \frac{2}{9}, \ and \ P(E_3) = \frac{1}{3}$$

Let,

E denote the event that 'Bonus Scheme' will be introduced.

$$P(E/E_1) = \frac{3}{10}, \quad P(E/E_2) = \frac{1}{2}, \ and \ P(E/E_3) = \frac{4}{5}$$

Bayes Theorem

$P(E) = P(\text{The Bonus Scheme will be introduced})$

$$P(E) = P(E_1 \cap E) + P(E_2 \cap E) + P(E_3 \cap E)$$

According to the multiplication theorem of probability, we have,

$$P(E) = P(E_1)P(E/E_1) + P(E_2)P(E/E_2) + P(E_3)P(E/E_3)$$

$$P(E) = \frac{23}{45}$$

Using the Bayes' theorem, the probability that 'Bonus Scheme' will be introduced by manager A:

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E)} = \frac{6}{23} = 0.26$$